ANALYZING AUTHORIZED PUSH PAYMENT (APP) FRAUD

Insights from a 12-Month Transaction Dataset
Duration: Jan–Dec (synthetic, modeled on UK patterns)

Nandit Nandan Paul
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BUSINESS CONTEXT AND OBJECTIVE

Problem: APP fraud remains one of the costliest forms of digital payment fraud in the UK.

Objective:

- Identify channels and countries most exposed to fraud
- Assess reimbursement performance
- Evaluate claim resolution efficiency

Goal: Derive actionable insights to strengthen fraud mitigation and customer protection.

DATA AND METHODS

Dataset: 80K transaction-level records (12 months)

Tech Stack: SQL (Bigquery), Power BI

Approach: Derive actionable insights to strengthen fraud mitigation and customer protection.

1. Data Cleaning & Validation in SQL
2. Exploratory Visualization in Power BI
3. Insight Generation → Fraud trends, channel risk, reimbursement rates

KEY INSIGHTS

- Online & Mobile Channels: Account for *95%* of all APP fraud cases.
- Fraud Peaks: Fraud spikes in March, May, and September, possibly linked to tax season transactions, spring/summer travel spending, and post-holiday payment surges, when fraudsters exploit higher transaction volumes and relaxed customer vigilance.
- Resolution Lag: Online fraud claims take *1.3× longer to close than in-person.*
- Reimbursement Gap: Only *~60%* of losses are reimbursed.
- High-Risk Destinations: *Few countries* show disproportionate fraud rate – targeted monitoring required.

RECOMMENDATIONS & ACTIONS

- 🧩 Channel Risk Mitigation: Strengthen verification for online/mobile transfers.
- 💰 Faster Resolution Framework: Automate claim triage and prioritize high-value cases.
- 🌍 Cross-border Oversight: Enhanced scrutiny on high-risk corridors.
- 📊 Continuous Monitoring: Establish fraud dashboards updated monthly.

THANK YOU